

LUANG PHABANG, Lao People's Democratic Republic

WMO: **489300**Lat: **19.897N**Lon: **102.161E**Elev: **291**StdP: **97.88**Time Zone: **7.00 (E07)**Period: **05-19**WBAN: **99999**

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	10.8	12.8	7.9	6.9	17.6	9.7	7.8	16.8	5.7	13.8	4.3	20.5	0.7	140	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
5	10.2	37.9	24.9	36.6	24.7	35.2	24.7	27.6	33.0	27.1	32.5	26.6	31.7	2.6	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
26.3	22.5	29.9	25.8	21.9	29.1	25.4	21.4	28.5	90.3	33.3	87.5	32.5	85.6	32.0	32.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 5.2	3.9	3.1	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(5)			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	26.2	21.0	23.8	26.0	28.8	29.1	28.7	28.3	28.0	28.0	26.8	24.1	21.5
	DBStd	3.46	2.77	2.11	2.92	2.10	2.01	1.53	1.47	1.24	1.21	1.55	2.41	2.39
	HDD10.0	1	1	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	18	11	1	2	0	0	0	0	0	0	0	0	5
	CDD10.0	5906	342	387	497	563	593	562	567	558	541	519	422	356
	CDD18.3	2882	93	155	241	313	334	312	309	299	291	261	172	102
	CDH23.3	26718	515	1343	2363	3476	3696	3275	3027	2772	2665	2058	1061	467
	CDH26.7	11477	136	619	1258	1881	1814	1378	1138	975	1022	813	344	97
(14) Wind	WSAvg	1.1	1.0	1.1	1.3	1.3	1.4	1.2	1.2	1.1	1.0	1.0	0.9	1.1
(15) Precipitation	PrecAvg	1560	15	13	48	96	172	229	308	343	235	80	29	23
	PrecMax	1907	72	76	156	182	281	511	423	536	482	205	117	149
	PrecMin	1116	0	0	0	8	64	102	120	180	122	17	0	0
	PrecStd	224	20	19	39	45	55	82	75	94	75	47	29	38
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	31.3	35.5	37.4	39.8	39.1	36.2	36.1	35.0	34.8	34.2	33.0	30.9
		MCWB	21.5	22.2	22.7	24.8	26.2	26.2	25.5	26.4	26.4	25.5	24.4	22.7
	2%	DB	29.9	34.0	36.2	38.3	37.6	35.1	34.5	33.6	33.9	33.2	31.5	29.0
		MCWB	21.2	22.0	22.9	24.8	25.9	25.9	25.9	26.2	26.3	25.2	23.9	21.7
	5%	DB	28.3	32.6	35.1	36.9	35.9	34.0	33.2	32.7	32.9	32.2	30.1	27.8
		MCWB	20.7	21.5	22.9	24.6	25.8	25.9	26.1	26.2	26.0	24.8	23.3	21.1
	10%	DB	26.8	31.0	33.8	35.2	34.2	32.8	32.0	31.5	31.9	31.1	28.8	26.2
		MCWB	20.0	21.2	22.7	24.4	25.7	25.9	25.9	26.0	25.8	24.5	22.5	20.5
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	23.7	25.1	25.8	27.7	28.3	28.2	27.6	28.2	27.9	27.2	25.9	23.7
		MCDB	29.1	32.2	33.6	34.7	35.8	32.7	32.9	32.4	32.7	32.0	30.5	28.5
	2%	WB	22.3	23.7	24.7	26.5	27.1	27.1	26.9	27.2	27.0	26.2	24.9	22.7
		MCDB	27.2	31.0	32.3	34.2	33.6	32.4	31.8	31.6	32.0	31.2	29.5	27.3
	5%	WB	21.4	22.8	24.0	25.7	26.5	26.5	26.4	26.6	26.5	25.6	24.3	21.9
		MCDB	25.9	29.9	31.8	33.1	32.5	31.6	30.9	30.9	31.1	30.3	28.5	26.0
	10%	WB	20.7	21.9	23.3	25.1	26.1	26.1	26.0	26.1	26.1	25.1	23.6	21.2
		MCDB	24.8	28.4	30.9	32.1	31.7	30.8	30.2	30.1	30.4	29.4	27.3	25.0
(35) Mean Daily Temperature Range	5% DB	MDBR	12.0	15.8	14.3	12.7	10.2	8.4	7.6	7.6	8.3	9.7	10.7	10.5
		MCDBR	14.3	18.0	17.0	15.2	12.6	10.3	9.8	9.4	9.8	10.6	11.1	11.8
		MCWBR	6.2	7.1	5.8	4.5	3.5	3.0	2.8	3.2	3.3	3.9	4.3	5.1
	5% WB	MCDBR	12.2	16.2	15.7	13.6	10.9	9.2	8.4	8.6	9.1	10.0	10.2	10.6
		MCWBR	6.1	7.4	6.2	4.8	3.7	3.1	3.0	3.4	3.5	4.2	4.3	5.3
(40) Clear-Sky Solar Irradiance	taub		0.431	0.538	0.789	0.749	0.493	0.446	0.444	0.472	0.494	0.526	0.428	0.421
	taud		2.213	1.919	1.435	1.549	2.193	2.357	2.354	2.255	2.174	2.044	2.275	2.254
	Ebn at Noon		838	764	601	637	815	847	850	831	810	765	836	834
	Edh at Noon		133	189	318	286	149	125	126	139	150	165	125	125
(44) All-Sky Solar Radiation	RadAvg		4.12	4.92	5.05	5.45	5.08	4.45	4.13	4.25	4.56	4.41	4.33	3.88
	RadStd		0.31	0.40	0.33	0.41	0.45	0.40	0.49	0.34	0.19	0.24	0.34	0.32

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (5 neighbors)		+0.27	N/A	N/A	+0.53	N/A	N/A	N/A	N/A	+154	+151

Nomenclature: See separate page